SAMPLE

Great Observatories: The FAST Telescope

China's Giant Radio Eye

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Introduction

Before the ear there was a valley, and the valley had been waiting a few hundred million years in the karst of Guizhou for someone to notice that it was already shaped like a bowl. The Five-hundred-meter Aperture Spherical Telescope, which the Chinese call Tianyan and everyone else translates as the Sky Eye, is the largest single dish ever built, and it listens where the rock already dipped. This volume is the ear's biography: how one researcher walked the karst for a decade with a camera and a notebook, how a nation approved the impossible budget, how four thousand panels learned to move, and how the ear repaid the trouble by finding more spinning stars in seven years than every other dish on Earth found together.

The ear's specialty is patience. A radio telescope gathers waves that arrive feeble after crossing the galaxy, and the only honest responses are a bigger bowl and a quieter valley. The Sky Eye chose both. Its dish sits inside a natural depression ringed by hills that shield the site from the telephones of the world, and its surface does not sit still: forty-four hundred panels on winches flex a moving patch of the surface into the focusing curve while the sky drifts overhead, so that the giant can look where it wants rather than only straight up. The instrument listens from ten centimeters to more than four meters of wavelength, a range that covers the longest whispers the galaxy makes.

The walk ahead has twelve chapters between two doors. The rooms hold the discovery, the anatomy, the physics of the listening, the craft of measurement, the parts that move, the telescope's neighbors in history, the campaigns it runs, the finds it has made, the honest difficulties, the plans for the next decades, the cabinet of curiosities, and the closing verdict. Every room keeps the same rules as the rest of this series: the facts are told plainly, the errors of the past are admitted rather than varnished, and the people who did the work keep their names.

The reader is invited to enter in order or by the door that interests them, and the ear will not mind either way. It has never once cared what anyone thought of it. It only listens, and the listening has been magnificent.

The reader should also know what these pages decline to be. They are no technical manual; the blueprints live in the journals and in the maintenance logs of the observatory, and no reader needs the load ratings of six cable towers to love them. It is not an institutional history either; the committees matter, but the committees do not walk the karst. What this book keeps is the story a curious visitor would want: the valley, the panels, the winches, the clocks in the sky, the founder's long patience, and the strange fact that the most sensitive ear on Earth sits in one of the quietest hollows on Earth, and both of those things had to be made true at once.

One number belongs here before the walk starts, because the number explains everything else. The dish hears signals that arrive with powers around a micro-jansky, which means the energy the ear gathers in a year of listening would not warm a spoon of soup. The universe is not loud. It is almost silent, and the instruments that study it are machines for hearing the almost. Every fact in the coming rooms, the winches, the panels, the surrendered phones, the maser clocks, exists in service of that almost, and when the reader understands the almost, the whole valley makes sense at once.

The series' other standing habit is declared here too, for the browsers who skim the introductions: each volume of Great Observatories tells the biography of one instrument, the way a shelf of lives tells a family, and the family in this case includes the ground and the sky together, since a radio telescope is half machine and half geography. The FAST volume keeps the balance: the people who chose the hollow, the engineers who filled it, and the clocks and gas it hears, all in the house style of this series, which is plain sentences, checked numbers, and no varnish on the hard parts. The valley gets the same respect the stars get, which is unusual for an astronomy book, and it is the only possible policy for a telescope that is, in the end, mostly hole.

The introduction ends with the customary honesty about scale: the reader will meet numbers here that strain intuition, five hundred meters of dish, thousands of panels, a sensitivity measured in fractions of a hot meal's warmth per year, and the series asks the reader to let the numbers pass through rather than stop at them, because every one of them was chosen to serve one purpose, and the purpose is hearing. The ear was built by people who thought the galaxy was worth straining intuition for, and this book was written on the same assumption, and the assumption is the only one the shelf keeps.

A word on the voice, since the reader has now heard enough of it to judge: this volume speaks of the telescope as an ear, and the metaphor is chosen because the instrument's own makers speak of it as an eye, and the two organs share the work between them, the eye naming the dish and the ear describing the work. Neither is wrong. The panels do look like an iris from the rim, and the data does arrive as sound in every sense that matters, a

whisper, a tick, a hum, and the reader who keeps both organs in mind will find every chapter of the biography perfectly embodied: an eye that listens, in a valley that watches, for a sky that has been talking all along.
The introduction has one more piece of housekeeping, the map of the rooms, and the map is short: the next three chapters carry the anatomy and the physics, the middle four carry the craft, the history, and the finds, the last four carry the difficulties, the plans, the curiosities, and the verdict, and the two doors at either end belong to the reader. Nothing in the middle requires anything before it, and everything in the middle rewards the order, and the shelf has arranged the rooms both ways at once, which is the shelf's own kind of quiet engineering, and the engineering works.

Chapter 1: The Discovery of The FAST Telescope

The discovery of the Sky Eye was a site hunt before it was an engineering project. Nan Rendong, a researcher at the Chinese Academy of Sciences, proposed the telescope in 1994, and the first problem was not money but geography: a five-hundred-meter dish cannot be built on flat land without digging a crater, so the project needed a crater nature had already dug. The karst country of Guizhou, in the south of China, is soft limestone riddled with sinkholes, and Nan spent more than a decade walking that country with colleagues, comparing depressions on foot. The team considered hundreds of candidates, and the winner was a sinkhole called Dawodang in Pingtang County, deep, round, drained, and flanked by three hills that framed a natural radio sanctuary.

The valley was chosen for three quiet reasons. Its shape was close enough to the sphere the designers wanted to spare them most of the excavation. Its floor drained through the karst, so rainwater would not drown the dish. And its remoteness was a gift no engineer could build: the nearest small city was far away, the hills blocked stray transmissions, and the site offered a radio silence that the country could then protect by law. When the National Development and Reform Commission approved the project in July 2007, the approval came with a radio-quiet zone around the valley, a fence of regulations several kilometers deep.

Construction began in March 2011, and the difficulties arrived on schedule. The site had poor road access, so the concrete and cable arrived slowly. The actuators under the panels emitted their own radio noise at first, a telescope whispering to itself, and they were redesigned and shielded until the ear could hear the sky instead of its own machinery. The budget grew from the planned seven hundred million yuan to about one point two billion, roughly one hundred eighty million dollars at the time, a sum that bought the largest single instrument on the planet for less than the cost of a single modest space mission. The last panel went in on the morning of 3 July 2016, and the valley was, for the first time, one machine.

First light came on 25 September 2016, and the ear opened slowly. The early observations ran without the active surface, holding the reflector in a fixed shape and letting the Earth's rotation carry the sky across the beam, the patient way. Full operation took three years of commissioning, and the telescope was declared formally operational on 11 January 2020. By then the founder was gone: Nan Rendong, chief scientist and chief engineer of the project for twenty-two years, died of lung cancer in Boston on 15 September 2017, ten days after the instrument's first confirmed discoveries were announced. He had walked the karst, built the ear, and heard it speak. The listening he started has not stopped since.

The proposal's own history deserves a paragraph, because the timing was not obvious in 1994. China's radio astronomy then ran on modest dishes and a determination to catch up, and a five-hundred-meter telescope was a paper dream at a scale the country had never attempted. Nan and his colleagues carried the concept through a decade of site studies and feasibility reports, through the international radio astronomy community's debates about whether the future belonged to dishes or to arrays, and into the national planning process, where the proposal survived on the strength of one argument: the largest collecting area hears the faintest things, and the faintest things are where the unknown lives. The 2007 approval was the hinge of the whole story, and the hinge had been oiled for thirteen years.

The commissioning years, 2016 to 2020, are the part of the discovery story the ribbon-cuttings skip.
The record of that phase is kept in the observing logs' own language, and the language is patient: the early scans ran without the deformable surface, the first pulsars were found in the drift, and each capability was declared ready only after months of proof. The team published the commissioning results in the trade's own journals, and the results read as a slow crescendo, sensitivity improving, receivers maturing, the noise floor dropping, until the formal declaration in January 2020 simply confirmed what the sky had been saying for a year. The ceremonies are dates; the discovery was a curve. The active surface had to be tuned until a three-hundred-meter patch held its parabola within millimeters; the cable robot had to learn to park the feed cabin over that patch while the wind

argued; the receivers were swapped, debugged, and in one case replaced by a better idea born during testing. The early science ran on patience, scanning whatever drifted overhead, and the patience paid before the machine was even finished: the first pulsars were found with a commissioning receiver built as a stopgap. The lesson of the commissioning years is the lesson of the whole project, and it is worth stating plainly: the ear's builders treated incompleteness as a phase, not an excuse, and the discoveries did not wait for the ceremony.

The chapter's last exhibit is a small one and the series keeps it anyway: the photograph of the last panel being set, on the July morning of 2016's third day, is the image the observatory uses on its posters, and the image shows workers in orange on a cable net above a green valley, a construction scene indistinguishable from a dozen other megaprojects until the viewer knows what the machine is for. The knowing is the biography's whole work, and the chapter has done its share of it, and the rooms that follow carry the anatomy.

One sentence of the founder's deserves quoting for its plainness, because the biography is fuller with his own words than with any paraphrase: Nan said of the site selection years that he had examined the candidate depressions one by one, on foot, and that the walking was the science. The sentence is kept here without decoration, as the shelf keeps such things, because it explains the twenty-two years better than any chronology can: the project advanced at the speed of a man in bad country, and the speed was the right one, and the ear proves it.

The chapter closes with the founding coincidence that the volume keeps because no novelist would dare it: the proposal that became the ear was first sketched in 1994, and the founder's twenty-two years ended within weeks of the first discoveries, and the instrument's most productive decade has been the one he did not see. The biography keeps that without comment, as the discipline keeps such facts, and the reader is free to keep it as they like, and the rooms ahead are the rooms he planned, and the planning is enough.